

Randomized Complete Block Design

A *randomized complete block design* (rcbd) is a restricted randomization design in which experimental units are first selected into homogeneous groups, called *blocks*, and the treatments are then assigned at random within each block.

How to randomize? The randomization procedure for a rcbd is straightforward. Within each block a random permutation is used to assign treatments for experimental units, just as in a completely randomized design (crd). Independent permutations are selected for different blocks.

Example: We are interested to determine whether or not four different tips produce different readings on a hardness testing machine. The machine operates by pressing the tip into a metal test coupon, and from the depth of the resulting depression, the hardness of the coupon can be determined. The experimenter has decided to obtain four observations for each tip. There is only one factor of interest here: tip type.

For a crd we need 16 different metal coupons, one for each run in the design. The problem with this design is that if the metal coupons differ slightly in their hardness, then the experimental units will contribute to the variability in the hardness data. As a result experimental error will reflect both the random error and the variability between coupons. We would like to make the experimental error as small as possible; that is we would like to remove the variability between coupons from the experimental error. A design that will accomplish this requires that the experimenter tests each tip once on each of the

coupons. The word "complete" indicates that each block (coupon) contains all the treatments (tips). The observed response is the Rockwell C scale minus 40.

The rcdb for the hardness testing experiment

Tip Type	Test Coupon			
	1	2	3	4
1	9.3	9.4	9.6	10.0
2	9.4	9.3	9.8	9.9
3	9.2	9.4	9.5	9.7
4	9.7	9.6	10.0	10.2

Remarks:

1. In a rcdb, each block constitutes a replication of the experiment. For that reason, it is highly desirable that the experimental units within each block be processed together whenever this will help to reduce experimental error variability. For example, an experimenter may tend to make changes in experimental techniques over time without being aware of it. Consecutive processing of the experimental units block by block will tend to keep such sources of variation out of the variation within blocks, and thereby make the experimental results more precise.

2. In factorial experiments, some of the factors of interest are often characteristics of the experimental units, such as gender, age and amount of experience on the job. Even though these factors are not introduced to reduce experimental error variability but rather for their intrinsic interest, we shall nevertheless consider such experiments as crbds, since the randomization of treatments to experimental units is restricted by nature of the classification factors considered.

Criteria for Blocking:

To help recognize some of the characteristics of experimental units that are fruitful criteria for blocking, we need a precise definition of an experimental unit. All elements of the experimental situation that are not included in the definition of the treatment need to be assigned to the definition of the experimental unit. Other elements of the experimental setting might be also included in the definition of the experimental unit, and should be if they could be the cause of variability in the observations. This suggests two types of criteria for blocking:

- a. Characteristics associated with the units - **for persons:** gender, age, income, intelligence, education, job experience, etc.; **for geographic areas:** population size, average income, average temperature, population density, type of area, etc.
- b. Characteristics associated with the experimental setting: observer, time of processing, machine, batch of material, measuring instrument, etc.

Advantage and disadvantages of the rcbd:

Advantages:

- a. With effective grouping it can provide substantially more precise results than the crd.
- b. It can accommodate any number of treatments and replications.
- c. Different treatments need not have equal sample size. Can always take multiple blocks of the same treatment (e.g. for a control).
- d. The statistical analysis is relatively simple.
- e. If needed can drop an entire block without complicating the analysis.
- f. Variability in experimental units can be deliberately introduced to widen the range of validity of the experimental results.

Disadvantages:

- a. Missing observations within a block require more complex calculations.
- b. The degrees of freedom for experimental error are smaller than in a crd. One df is lost for each block after the first one.
- c. More assumptions are needed for the model (e.g., no interaction between treatment and blocks, constant variance from block to block).

d. It is a more complicated design (treatment effect is fixed or random, blocking effect is fixed or random).

Model:
$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \begin{matrix} i=1,\dots,a \\ j=1,\dots,b. \end{matrix}$$

If we assume that the treatment levels are fixed and the block levels are fixed then we assume that:

$$\sum_{i=1}^a \tau_i = 0 \quad \text{and} \quad \sum_{j=1}^b \beta_j = 0.$$

Quite often the block levels will be random, then instead of assuming that the sum of the block effects add up to 0, we will assume that:

$$\beta_j \text{ are iid } N(0, \sigma_b^2).$$

We also assume that

$$\varepsilon_{ij} \text{ are iid } N(0, \sigma_e^2)$$

and are independent of β_j 's if they are random.

We are interested in testing the hypotheses (blocks fixed):

$$H_0 : \mu_1 = \dots = \mu_a \text{ vs. } H_1 : \text{not all } \mu_i \text{'s are equal,}$$

where

$$\mu_i = \frac{1}{b} \sum_{j=1}^b (\mu + \tau_i + \beta_j) = \mu + \tau_i.$$

The above hypotheses are equivalent to:

$$H_0 : \tau_1 = \dots = \tau_a = 0 \text{ vs. } H_1 : \text{not all } \tau_i\text{'s} = 0.$$

Remark: If the block levels are random than we replace in the hypotheses above μ_i by $E(\mu_i)$. The second representation of the hypotheses remains valid, since the expected value of the i th treatment mean is equal to $\mu + \tau_i$.

The ANOVA identity for the rcdb:

$$\begin{aligned} \sum \sum (Y_{ij} - \bar{Y}_{..})^2 &= \sum \sum (\bar{Y}_{i.} - \bar{Y}_{..})^2 + \sum \sum (\bar{Y}_{.j} - \bar{Y}_{..})^2 \\ &\quad + \sum \sum (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2 \end{aligned}$$

or

$$SS_{\text{total}} = SS_{\text{treatment}} + SS_{\text{block}} + SS_{\text{error}}.$$

The breakdown of degrees of freedom is:

$$(ab - 1) = (a - 1) + (b - 1) + (a - 1)(b - 1).$$

One can show that the sums of squares on the right-hand side of the ANOVA identity divided by σ_e^2 have independent chi-square distributions with the degrees of freedom associated with the sums of squares. Therefore, their ratios have an F distribution.

ANOVA for a rcbd

Source of Variation	Sum of Squares	df	MS	F
Treatments	$\sum_{i=1}^b \{Y_{i.}^2\} - Y^2_{..}/N$	$a-1$	$SS_{\text{treatment}}/(a-1)$	$\frac{MS_{\text{treatment}}}{MS_{\text{error}}}$
Blocks	$\sum_{j=1}^a \{Y_{.j}^2\} - Y^2_{..}/N$	$b-1$	$SS_{\text{block}}/(b-1)$	
Error	by subtraction	$(a-1)(b-1)$	$SS_{\text{error}}/(a-1)(b-1)$	
Total	$\sum \sum Y_{ij}^2 - Y^2_{..}/N$	$ab-1$		

Model Adequacy for a Randomized Complete Block Design

One should be alert to potential problems with the normality assumption, unequal error variance by treatment or block, and block treatment interaction. Like in the crd design, residual analysis is the major tool in this diagnostic checking. The residuals for a randomized block design are

$$\begin{aligned} e_{ij} &= Y_{ij} - \hat{Y}_{ij} \\ &= Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..} \end{aligned}$$

A normal probability plot of the residuals should be examined along with the dot plots of the residuals or Y_{ij} 's vs the treatments and the blocks. Also a plot of the residuals vs the fitted values should be examined. We will use PROC GLM in SAS to produce these plots.

What are the conclusions from the above?

Remark: Sometimes the plot of the residuals vs the fitted values has a curvilinear shape; for example, there may be a tendency for negative residuals to occur with low fitted values, positive residuals with intermediate fitted values, and negative residuals with high fitted values (or the other way around). This type of pattern suggests that there exists a treatment block interaction. One can actually perform a statistical test to detect this interaction. We will discuss this test later. One can try to transform the data to minimize this interaction. For example, sometimes the following model makes sense:

$$Y_{ij} = \mu\tau_i\beta_j + \varepsilon_{ij}.$$

In this case the following transformation makes sense:

$$\ln Y_{ij} = \ln \mu + \ln \tau_i + \ln \beta_j + \varepsilon_{ij}.$$

Relative Efficiency of the rcbd

It is often useful to estimate the relative efficiency of the rcbd compared to a crd. One way to define this relative efficiency is:

$$R = \frac{(df_b + 1)(df_r + 3)}{(df_b + 3)(df_r + 1)} \times \frac{\sigma_r^2}{\sigma_b^2},$$

where σ_r^2 and σ_b^2 are the experimental error variances for the crd and rcbd, respectively and df_r and df_b are the corresponding error degrees of freedom. This quantity can be viewed as the increase in replication that is necessary if a crd is used as compared to a rcbd, in order to achieve the same sensitivity with the two designs. The ratio of degrees of freedom in R is an adjustment to reflect the different values of df for error in the two designs. To compute R we have to estimate the error variances by their MS for error from the two designs. We use the MS_{error} from the rcbd to estimate σ_b^2 . One can show (Cochran and Cox 1957) that:

$$\sigma_r^2 = [(b-1)MS_{\text{block}} + b(a-1)MS_{\text{error}}]/(N-1),$$

is an unbiased estimator of σ_r^2 . For our example we get that $R = 6.71$. Clearly blocking was quite effective in this experiment. The general rule of thumb is: if in doubt, then one should always block.

Estimating Missing Values in a rcbd.

When using a crbd sometimes an observation in one of the blocks is missing. This can happen because of carelessness or for reasons beyond the control of the experimenter. A missing observations introduces a new problem in to the analysis, since the treatments are no longer orthogonal to blocks. There are two general approaches to the missing value problem.

a. The missing value is estimated so that the SS_{error} as a function of the missing observation is minimized. Then the observation is inserted for the missing value and the ANOVA is performed just as if the missing value was the real data, with the error degrees of freedom reduced by one.

b. Fitting a regression model to the data where this observation is missing. PROC GLM will do it for us.

Using approach a. one can show that the missing observation y_{ij} is given by:

$$y_{ij} = (aY'_{i.} + bY'_{.j} - Y'_{..}) / (a-1)(b-1)$$

where $Y'_{i.}$, $Y'_{.j}$ and $Y'_{..}$ are the totals based on the data with the missing observation.

**Approximate ANOVA for the Hardness Data
with one Missing Value for tip = 2 and coupon = 3**

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Tip Type	.3998	3	.1333	17.12
Coupons	.7953	3	.2651	
Error	.622	8	.0078	
Total	1.2573	14		

The missing observation y_{23} has been estimated via the equation described above:

$$y_{23} = [4(29.1) + 4(28.6) - 144] / 9 = 9.6444.$$

Another approach, that we will use is to generate via

PROC GLM an estimate for this missing value.